Project Plan

FEM_BallisticDiffusion_PINN

Single living roadmap for ballistic-diffusive heat transport, superconducting-qubit quasiparticles, FEM implementation, and PINN acceleration

Purpose. This document is the top-level roadmap for **FEM_BallisticDiffusion_PINN**. The project builds a MATLAB-based finite-element simulator for nanoscale and microscale heat/quasiparticle transport in a superconducting-qubit-like geometry, then develops a physics-informed neural-network surrogate to accelerate geometry and material sweeps. The long-term design objective is to compare trap, heat-sink, pad, and material choices by how effectively they reduce quasiparticle density near the Josephson-junction-sensitive region.

Contents

Current architecture revision. This first project build uses a planar transmon geometry as the baseline physical qubit structure. The Josephson tunnel barrier is not directly meshed at nanometer thickness; it is represented by an effective junction-sensitive volume and source/sink terms. This is intentional because the substrate and capacitor pads are millimeter-to-micrometer scale, while the oxide barrier is nanometer scale. Directly resolving all scales in one first-pass 3D mesh would be inefficient and physically unnecessary for the first thermal/quasiparticle transport study.

1 Project objective

The project objective is to build a physics-first computational framework for studying how geometry, material choice, and engineered sinks influence quasiparticle poisoning in superconducting qubit structures. The project is not a full quantum-circuit simulator. Instead, it focuses on transport physics: how thermal energy, phonons, quasiparticles, and engineered traps move or remove nonequilibrium energy from regions that matter for qubit operation.

The intended physics chain is:

1. energy is deposited in a nanoscale or microscale region by a thermal, phonon, photon, or future radiation-like source,
2. thermal carriers propagate in a regime that may not be purely Fourier-diffusive,
3. low-temperature superconducting material properties modify heat capacity, thermal conductivity, quasiparticle density, diffusion, recombination, and trapping,
4. a realistic planar transmon geometry defines the sensitive junction region and possible locations for traps or heat sinks,
5. finite-element simulations generate high-fidelity solutions for selected designs,
6. a physics-informed neural-network surrogate learns the design-to-solution map so that many geometry and material designs can be swept more rapidly.

2 Module roadmap

No.	Module	Purpose

1	Ballistic-diffusive heat conduction at room temperature	Derive and implement the ballistic-diffusive heat-conduction equation from the Boltzmann transport equation under the relaxation-time approximation. The current priority is physics understanding: the note includes a detailed thin-film benchmark derivation, nondimensional form, finite-arrival-time ballistic flux, and slide-ready summaries.
2	Low-temperature superconducting material physics	Explain how the physics changes when the device is cooled below the superconducting transition temperature. Introduce the Bardeen-Cooper-Schrieffer energy gap, equilibrium and nonequilibrium quasiparticle density, superconducting heat capacity, quasiparticle diffusion, recombination, and trapping.
3	Three-dimensional planar transmon geometry and FEM implementation	Hard-code a 3D planar transmon-like geometry in MATLAB, including substrate, capacitor pads, effective Josephson-junction region, normal-metal quasiparticle traps, and optional heat-sink patches. Assemble scalar thermal and quasiparticle diffusion-reaction FEM systems.
4	Physics-informed neural-network acceleration	Build a physics-informed surrogate that is trained from FEM data and constrained by PDE residuals, boundary conditions, initial conditions, and positivity/energy constraints. Use it to accelerate geometry, trap, and material sweeps.
5	Future nonequilibrium electron-phonon and photon physics	Document future extensions: photon absorption, pair breaking, hot electron relaxation, high-energy phonon generation, quasiparticle recombination, phonon escape, and coupled Rothwarf-Taylor-type nonequilibrium dynamics. No code is required in the first build.

3 Baseline qubit geometry

The baseline geometry is a planar transmon fabricated as thin superconducting metal on a dielectric substrate. The model resolves the substrate, metal pads, trap regions, and effective junction-sensitive volume. It does not attempt to resolve the oxide barrier thickness.

Component	Baseline material	Baseline dimensions	Role
Substrate	Sapphire or high-resistivity silicon	3 mm $\times$ 3 mm $\times$ 0.5 mm	Main phonon and thermal body.
Left capacitor pad	Al, Ta, or Nb film	600 μm $\times$ 300 μm $\times$ 100 nm	One side of the transmon shunt capacitance and a superconducting transport region.
Right capacitor pad	Al, Ta, or Nb film	600 μm $\times$ 300 μm $\times$ 100 nm	Opposite shunt pad and superconducting transport region.
Pad gap	Exposed substrate/vacuum	40–80 μm	Capacitive electric-field region between pads.
Junction leads	Al	5–10 μm wide, 20–60 μm long	Connect pads to the effective junction-sensitive volume.
Josephson junction region	Effective Al/AlOx/Al region	0.2–1 μm lateral effective patch	Region in which quasiparticle density is most damaging.
Normal-metal traps	Cu, Au, or Pd	10 μm $\times$ 100 μm $\times$ 50–100 nm	Engineered quasiparticle sinks.
Backside heat sink	Cu, Au, or thick normal metal	200–500 μm lateral patch	Optional phonon/thermal sink on the substrate backside.

4 Run-order workflow

1. Compile and read the project plan and all physics notes.
2. Run `matlab/main_module1_bd_1d_validation.m` to validate the ballistic-diffusive thin-film model.
3. Run `matlab/main_module3_transmon_3d_fem.m` to build the baseline geometry, assemble the 3D FEM matrices, and solve an initial quasiparticle diffusion-reaction problem.
4. Run `matlab/cases/case_trap_distance_sweep.m` to compare no-trap, near-trap, and far-trap designs.
5. Use `matlab/main_module4_train_pinn_surrogate.m` only after enough FEM cases have been generated to train a surrogate meaningfully.

5 Design metric

The first optimization metric should target the region that is physically most sensitive, not merely the average chip temperature. A useful first metric is

$$J_{\mathrm{JJ}} = \int_0^{t_f} \int_{\Omega_{\mathrm{JJ}}} n_{\mathrm{qp}}(\boldsymbol{r}, t)\,\mathrm{d}V\,\mathrm{d}t, \tag{5.1}$$

where $n_{\mathrm{qp}}(\boldsymbol{r}, t)$ is quasiparticle density and Ω_{JJ} is the effective Josephson-junction-sensitive region. Lower J_{JJ} means fewer quasiparticles are present near the active weak link over the simulation window.

A geometry-regularized design objective can then be written as

$$J = J_{\mathrm{JJ}} + \lambda_V V_{\mathrm{trap}} + \lambda_G P_{\mathrm{geometry}}, \tag{5.2}$$

where V_{trap} penalizes excessive normal-metal trap volume and P_{geometry} penalizes fabrication-unfriendly geometries.

6 File map

Path	**Purpose**
`project_plan.tex/pdf`	Single living roadmap for project architecture.
`docs/shared/project_style.tex`	Shared LaTeX style file inherited from the RadiationEffects project style.
`docs/physics_notes/`	Textbook-style physics notes with purpose box, table of contents, symbols/acronyms section, derivations, and references.
`matlab/src/geometry/`	Geometry parameters and hard-coded cuboid representation of the planar transmon.
`matlab/src/materials/`	Material property models for substrate, superconducting films, and traps.
`matlab/src/module1_bd/`	Ballistic-diffusive thin-film validation code.
`matlab/src/module2_superconducting/`	BCS gap, equilibrium quasiparticle density, diffusivity, recombination, and trap-rate helper functions.
`matlab/src/module3_fem/`	3D scalar FEM assembly for thermal and quasiparticle diffusion-reaction equations.
`matlab/src/module4_pinn/`	PINN/surrogate model definitions, loss functions, and training skeleton.
`outputs/`	Generated figures, data, and case summaries.

7 Implementation philosophy

The project should remain physics-interpretable. Every MATLAB routine should map to a documented equation or modeling assumption in the physics notes. The first build is intentionally conservative:

- validate the reduced ballistic-diffusive equation before using it in complex geometry,
- treat the Josephson junction as an effective sensitive region rather than trying to mesh a nanometer oxide,
- use FEM data plus PDE residuals for the first PINN rather than a pure PINN trained without high-fidelity data,
- make all geometry and material parameters explicit and sweepable.

8 References

[1] G. Chen, "Ballistic-Diffusive Heat-Conduction Equations," *Physical Review Letters* **86**, 2297–2300 (2001).

[2] G. Chen, "Ballistic-Diffusive Equations for Transient Heat Conduction From Nano to Macroscales," *Journal of Heat Transfer* **124**, 320–328 (2002).

[3] J. Bardeen, L. N. Cooper, and J. R. Schrieffer, "Theory of Superconductivity," *Physical Review* **108**, 1175–1204 (1957).

[4] J. Koch et al., "Charge-insensitive qubit design derived from the Cooper pair box," *Physical Review A* **76**, 042319 (2007).